Optimization Baseline

M5 Retail Demand Forecasting — Problem Statement 11

Phase 1 — reproducibility and starting diagnostics

NPN AIA Hackathon — St. Joseph's College of Engineering

Phase 1 — re-running the current best model unchanged, to prove it reproduces before anything is optimised. Generated 2026-08-14.

> ***Terms. RMSE** — average error with big misses punished far more heavily; lower is better. **MAE** — the plain average error. **Leakage** — letting the model see information that would not have existed when the forecast was really made. **Tweedie** — a loss function for data that is never negative and mostly zero. **Objective / loss function** — the definition of "wrong" that the model tries to minimise. **Inner window** — an earlier 28-day period we tune on, so the real scoring window stays untouched.*

> *Every number here comes from an experiment that actually ran. Failed experiments are reported alongside successful ones — in this campaign most of them failed, and that is the finding.*

Why this step exists

Before changing anything we re-ran the existing best configuration from scratch. If it does not reproduce its own score exactly, then every later comparison is measuring randomness rather than our changes.

Result

	Reference	Re-run	Drift
RMSE	2.121043	2.121043	+0.00e+00
MAE	1.031927	1.031927	+0.00e+00

Reproduced exactly. Same seed, same data, same answer to every decimal place. The pipeline is deterministic.

Baseline behaviour we will try to improve

Diagnostic	Value	What it means
High-volume RMSE	5.9756	Error on the busiest 7.7% of rows
High-volume bias	-0.3891	Negative = we under-predict busy days
Share of squared error from that tier	61.33%	Where RMSE actually comes from
Mean prediction on true-zero rows	0.58485	We place a small positive value on days that turn out empty
Mean prediction where sales happened	2.31342	Against an actual mean of about 3.17
Prediction spread (p50 / p99 / max)	0.61136 / 12.44237 / 151.1688	The forecast is heavily concentrated near zero

The picture is consistent: the model is cautious. It under-predicts busy days and puts a little weight on quiet ones. Squared error punishes the first far more than the second, which is why over 60% of RMSE sits in a small minority of rows.

Leakage check

The extended feature builder used in later phases was put through the same corruption test: every sales value after the forecast origin was overwritten with 9999 and all features rebuilt. All 46 came back identical. The guarantee is re-earned, not inherited.

Next

Phase 2 adds candidate features one group at a time.